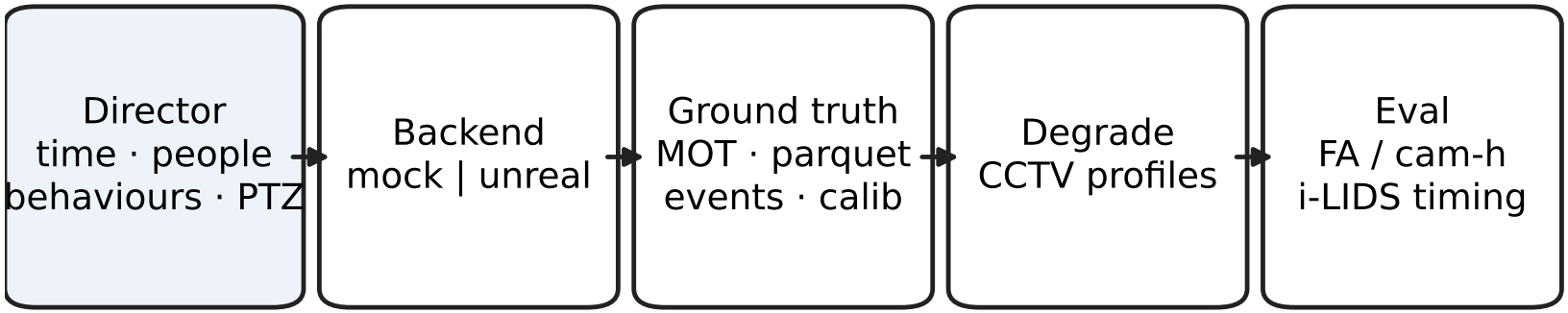
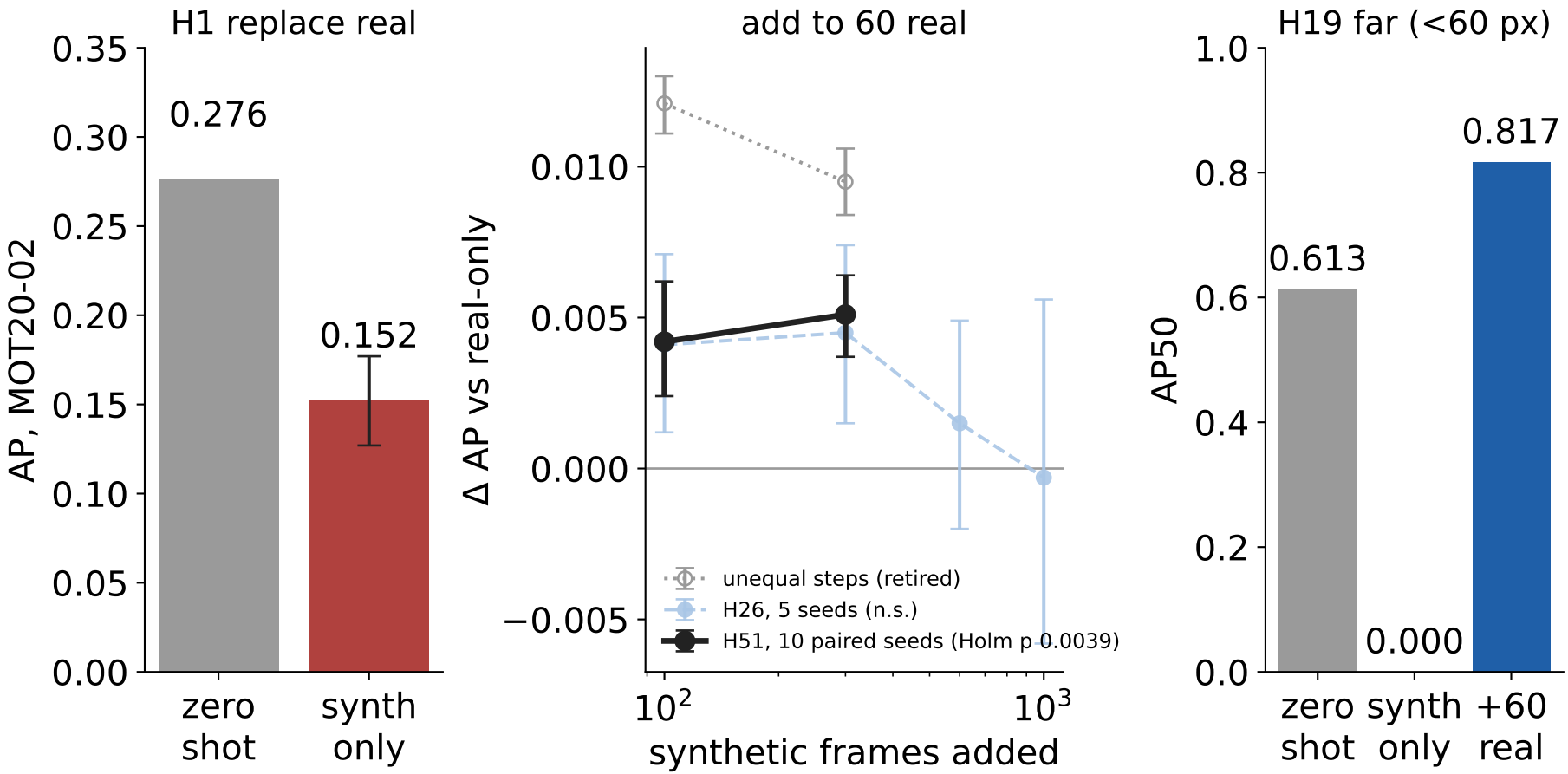
1 · SurvSim pipeline



engine-agnostic above the backend: SI units, seeded RNG, one definition of 'visible'

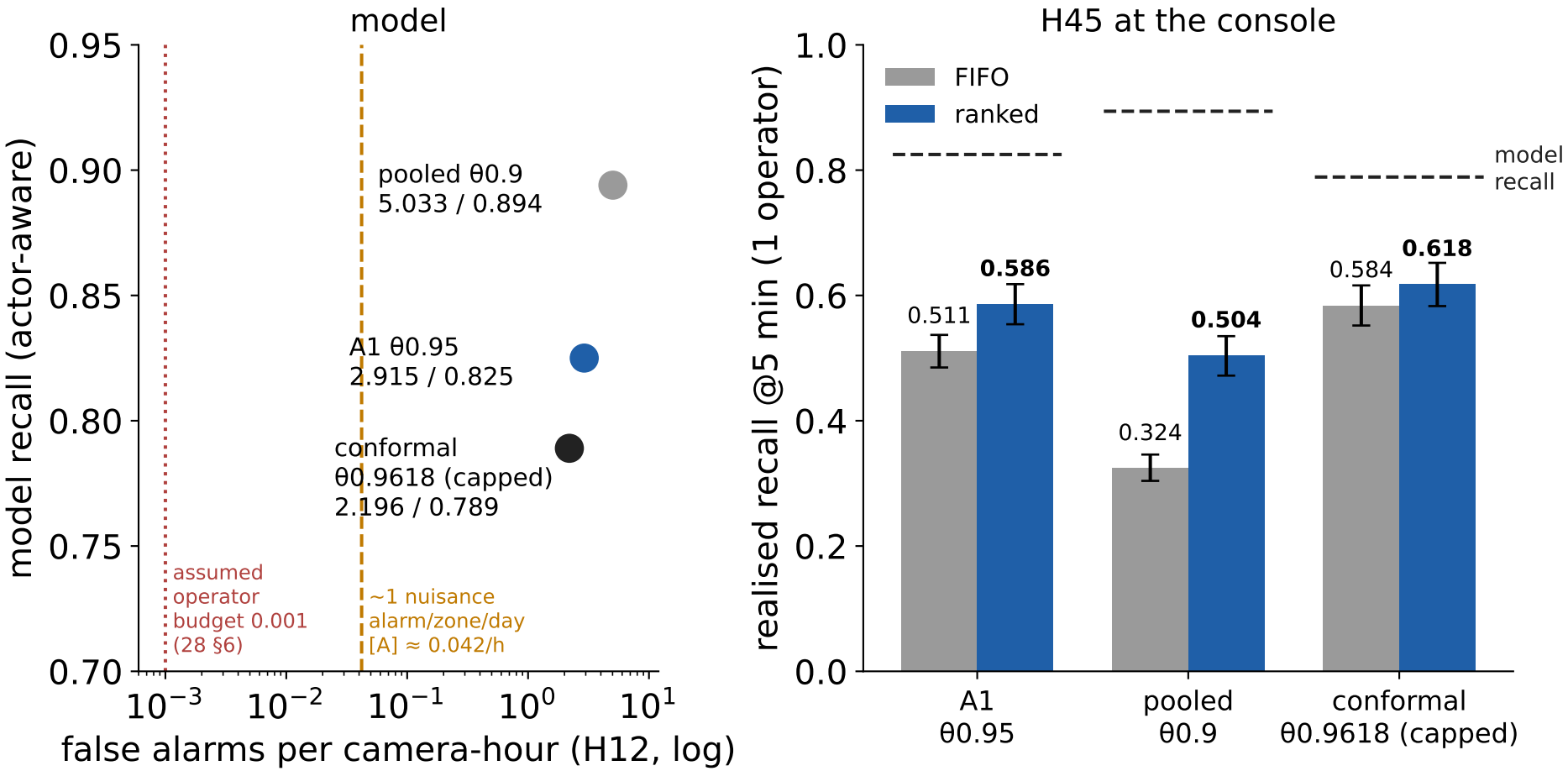
Architecture per CLAUDE.md / docs/02. Truth is recorded after avoidance and boxes come from rendered masks (occlusion included).

2 · Sim-to-real: small but significant add-on gain (H51)



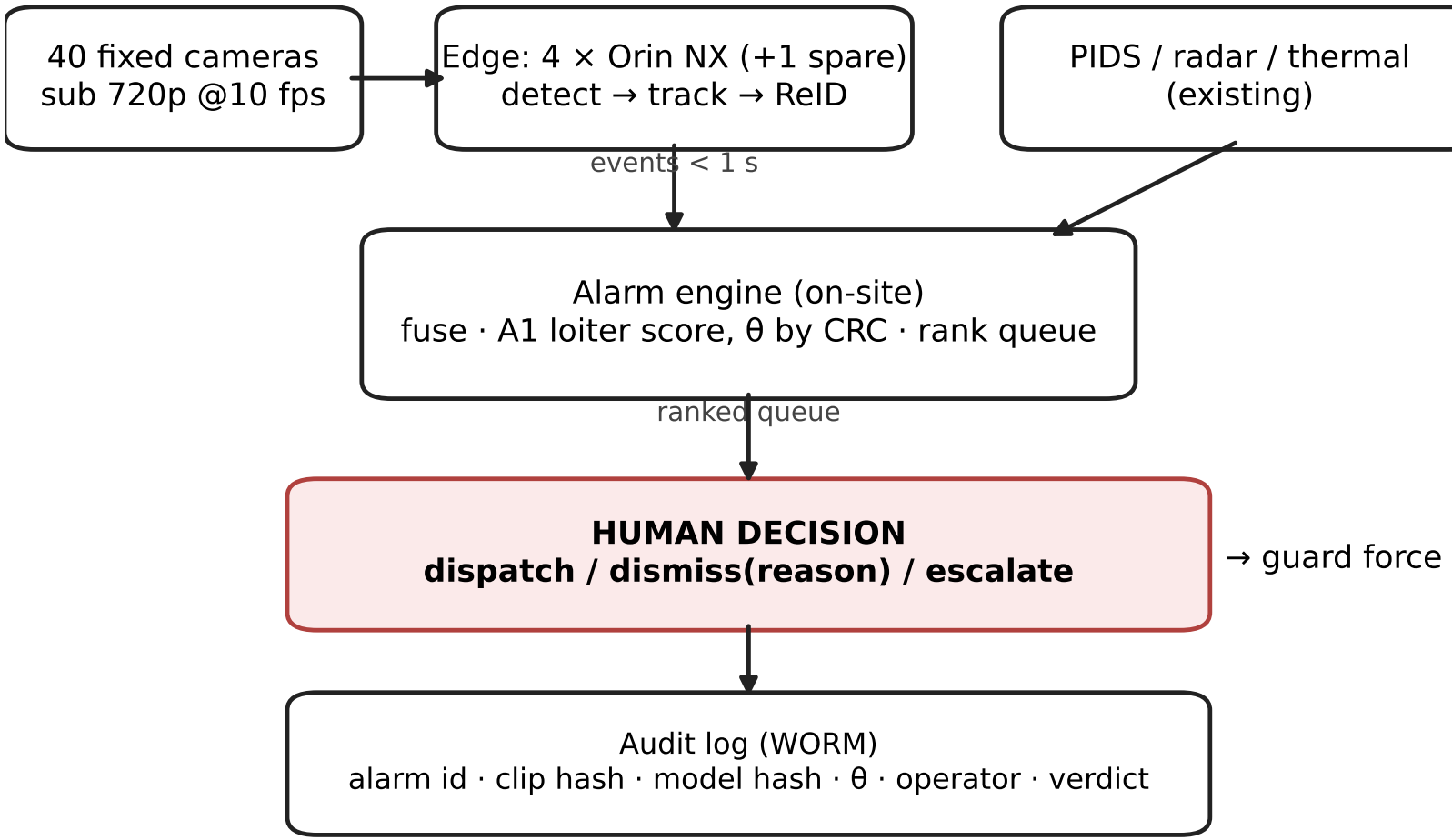
H1: synthetic-only 0.276 → 0.152 AP, MOT20-02. H7 (2 seeds) / h14b (5 seeds):
synthetic added to 60 real at UNEQUAL steps: mix_100 +0.0121, mix_300 +0.0095 over 6-epoch real-only 0.4862 (grey, retired label).
H51 (black, 10 paired seeds, 450 steps every arm, H26 recipe exactly): real-only 0.4911 [0.4895, 0.4926]; +100 0.4953 (gain +0.0042 [+0.0024, +0.0062]), +300 0.4962 (gain +0.0051 [+0.0037, +0.0064]), exact sign-flip Holm p 0.0039 both.
H26 (light blue, 5 seeds, superseded by H51): +0.0041 / +0.0045 / +0.0015 / −0.0003 at 100/300/600/1000 over real-only 0.4911; none Holm-significant.
H19: far synthetic-only 0.000 vs zero-shot 0.613; +60 real frames → 0.817 AP50.

3 · Loiter operating points: staffing picks the threshold



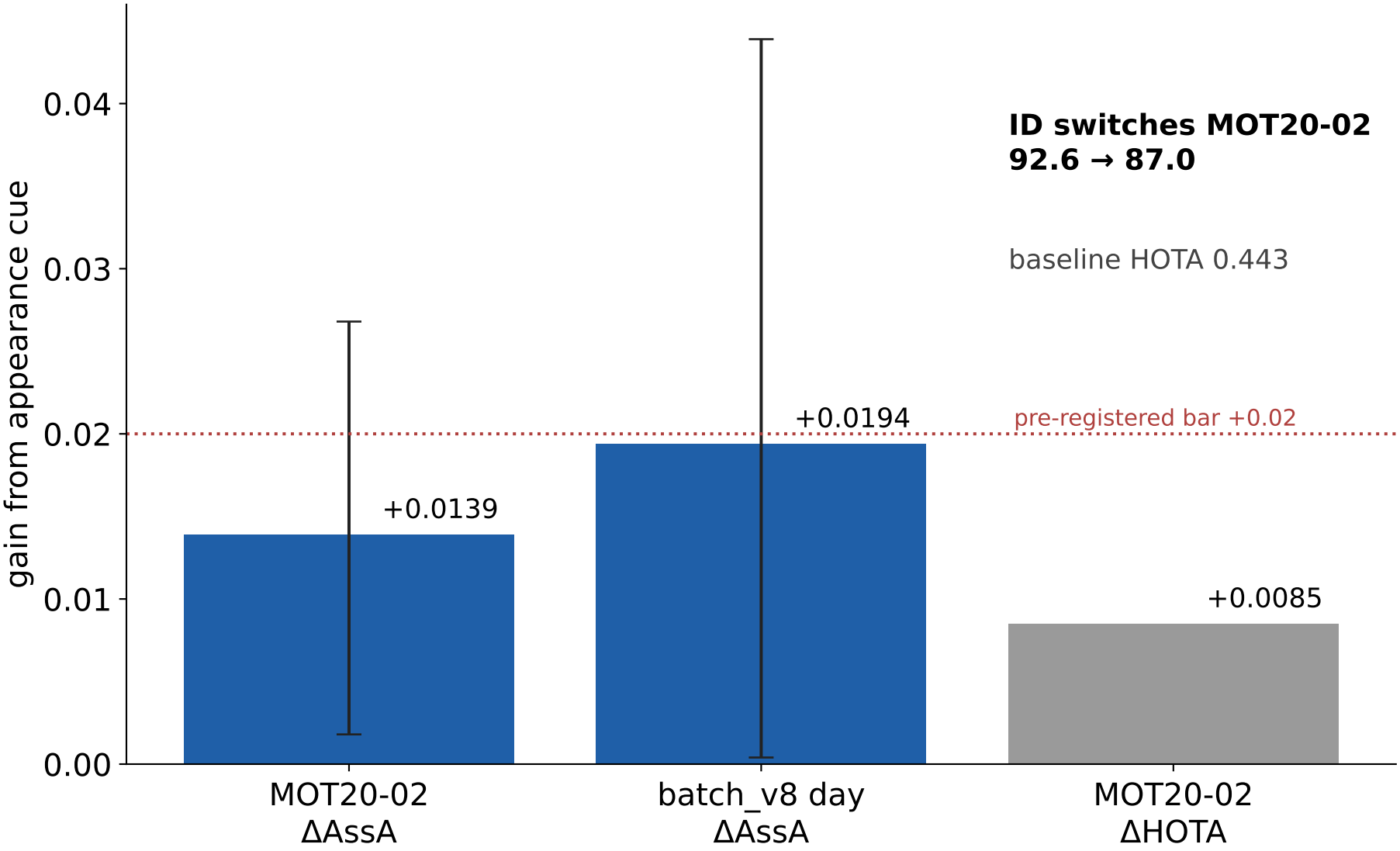
Left: model points on H12 (58 §8.4 / H42 / H33 / H33c): A1 θ0.95 2.915 / 0.825; pooled θ0.9 5.033 / 0.894; conformal (capped) θ0.961780 2.196 / 0.789 (CRC range 2.10–2.22). Our assumed operator budget 0.001 FA/cam-h (28 §6, A = 4 alarms/shift) is infeasible (H33 floor 0.248–0.263); external anchor ~1 nuisance alarm/zone/day [A] ≈ 0.042/h.
Right: H45c realised recall @5 min, 1 operator / 40 cams, 10 seeds, invented operator (H22): FIFO / ranked A1 0.511 / 0.586, pooled 0.324 / 0.504, conformal (capped) 0.584 / 0.618; loads 120 / 205 / 90.6 alarms/h. Ship: 1 operator → conformal (capped, exact B) ranked; ≥ 2 → A1 ranked; never pooled. Cap keys on GT identity; ID switches would reopen it (H59).

4 · Alarm path, 40-camera perimeter (74 drill 1)



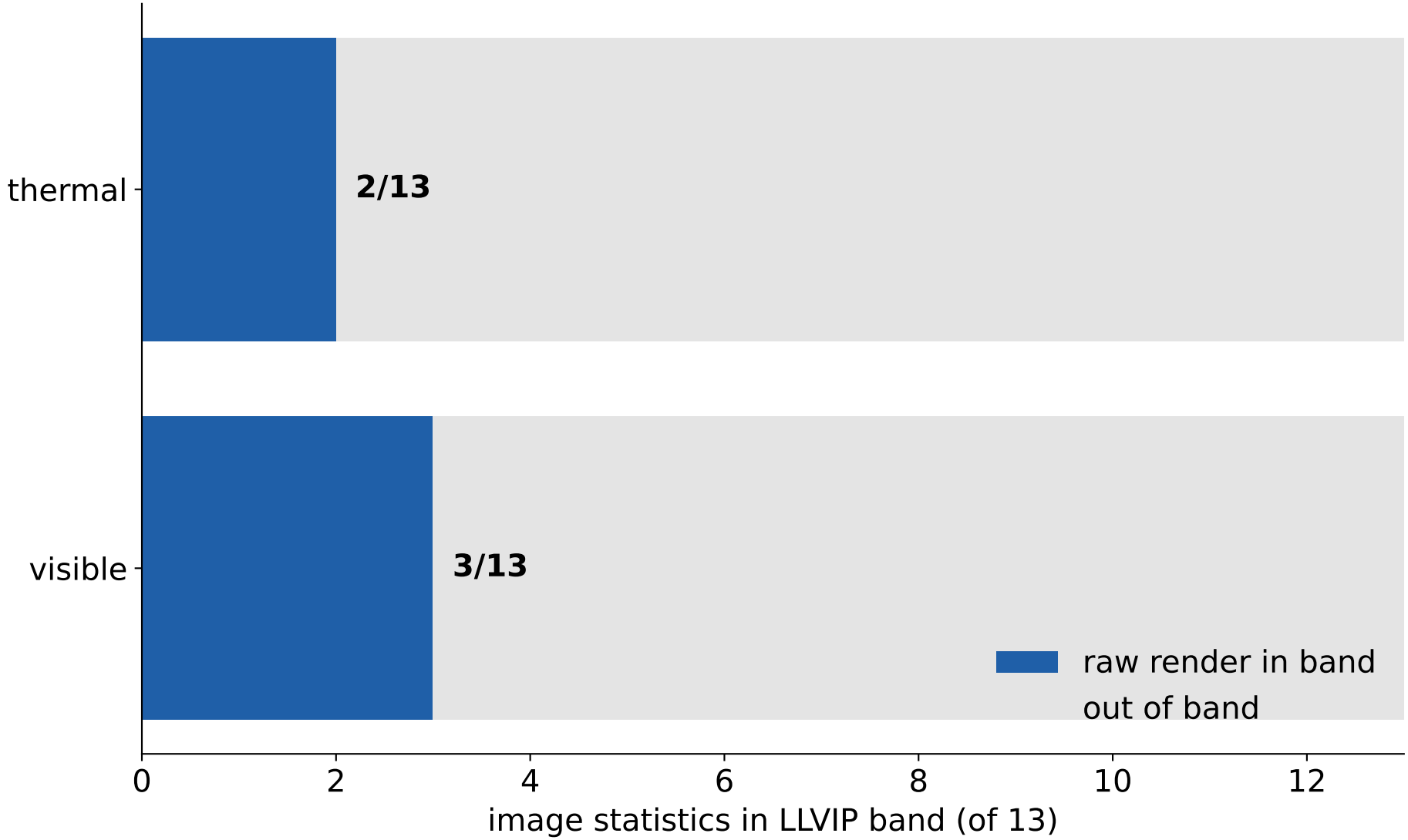
Edge sizing: 10–12 cams per Orin NX with detect + ReID (18× 1080p30 decode is a ceiling). Nothing is dispatched without the operator's click; every alarm and verdict is in a WORM audit log.

5 · Tracking: an appearance model helps association



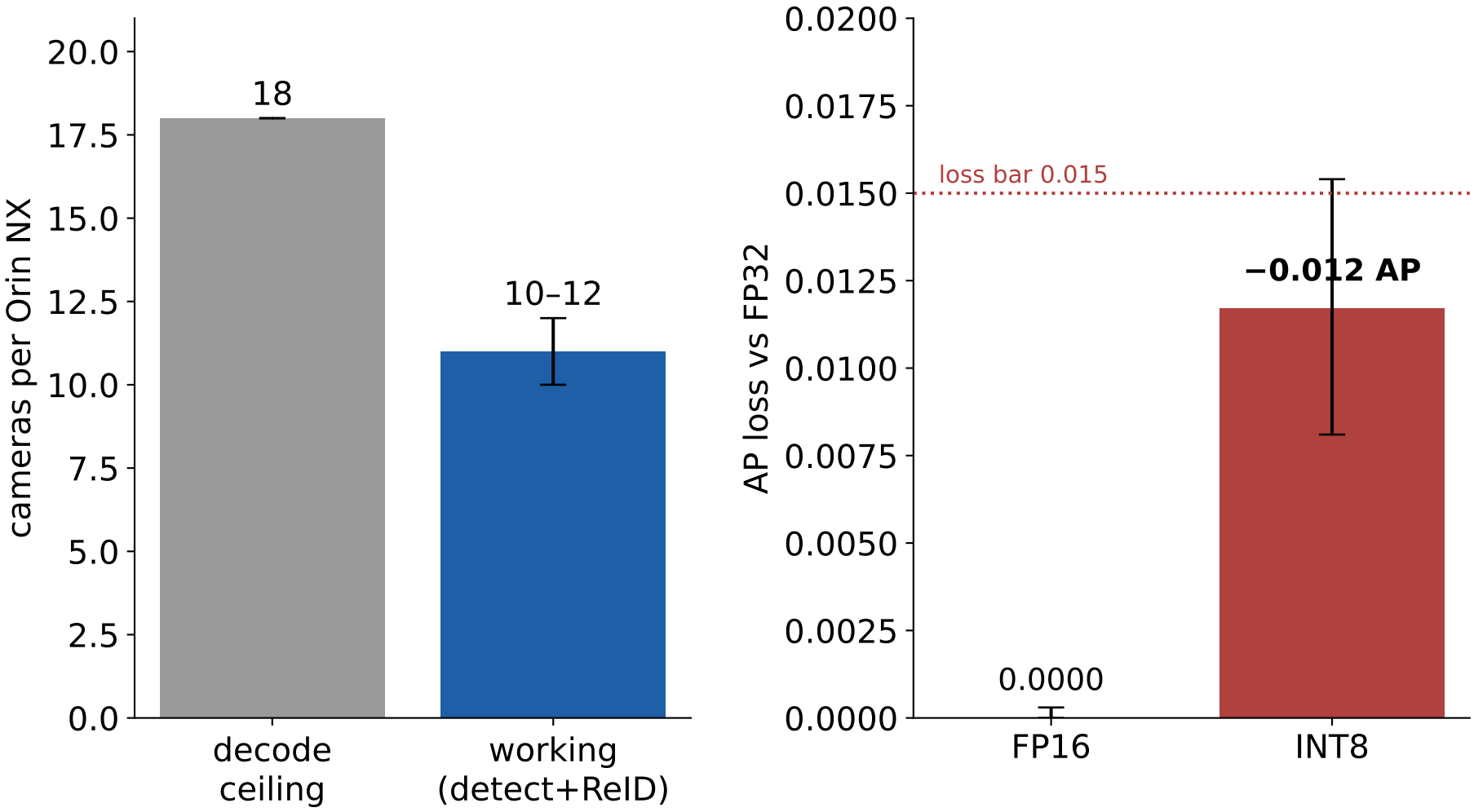
MOT20 subset HOTA 0.443 (TrackEval parity). H24: OSNet in ByteTrack stage 1, ΔAssA MOT20-02 +0.0139, batch_v8 +0.0194 (bar +0.02 not met); IDSW 92.6 → 87.0 (−5.6 [−11.0, −0.2]).
GT-box arm +0.0137 says association, not detection; synthetic *data* (H17) did not lift AssA.

6 · Night: the renders are not night-realistic yet



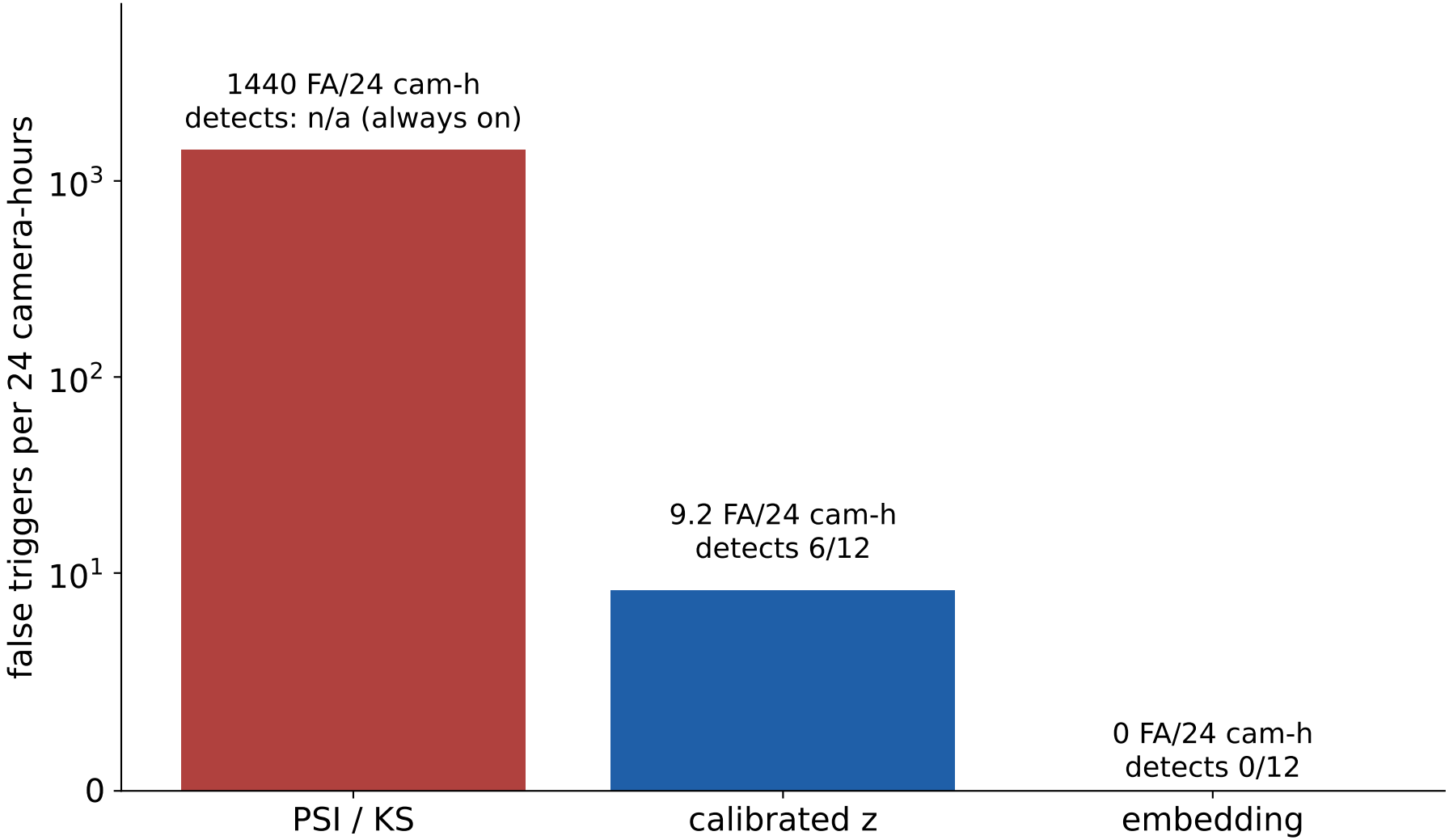
H32a: raw renders 3/13 visible · 2/13 thermal statistics in band vs LLVIP; tuned profiles overfit (9/13 → 2/13, 10/13 → 3/13 hold-out).
H32b: with no profile in band, the recall drops (IR −0.565, lowlight −0.835, thermal −0.470) measure the degrade model, so no recall is a night-penalty estimate.

7 · Edge budget: decode is a ceiling, INT8 is not free



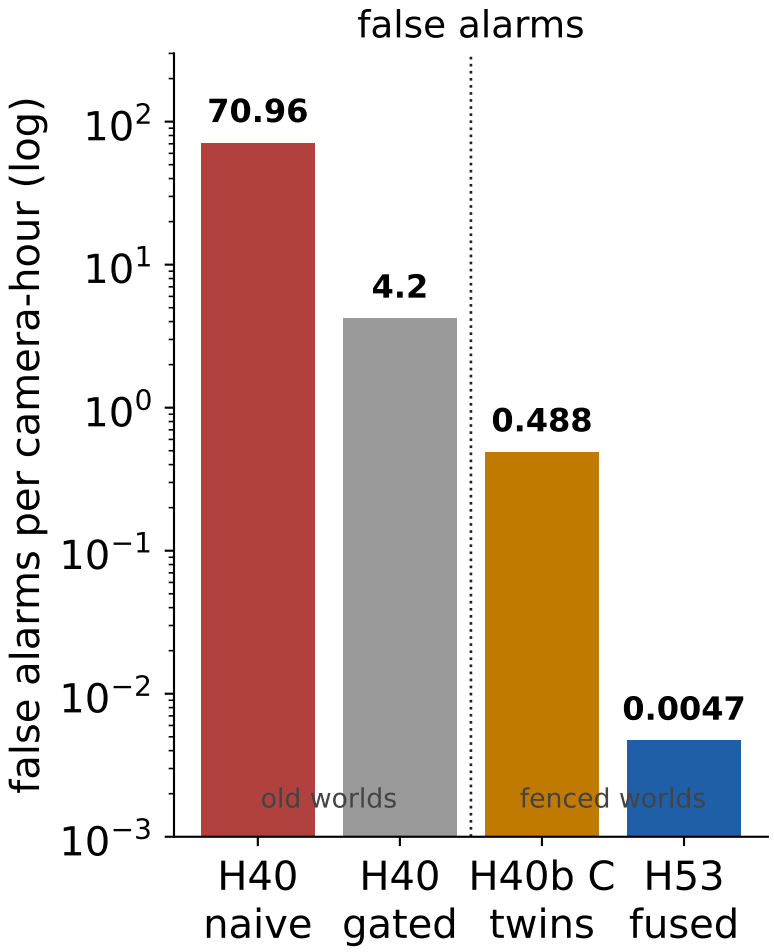
Orin NX 16 GB: 18× 1080p30 H.265 decode is a ceiling; ~10–12 cams/NX for detect 640 @10 fps + ReID (plan figure, ReID scaling [A]) → 4 NX for 40 cams.
H38 (yolov8n, MOT20-02 AP, 3 seeds): FP16 loss 0.0000; INT8 static PTQ loss 0.0117 (range 0.0081..0.0154; bar 0.015 not met, AP75 −0.030).

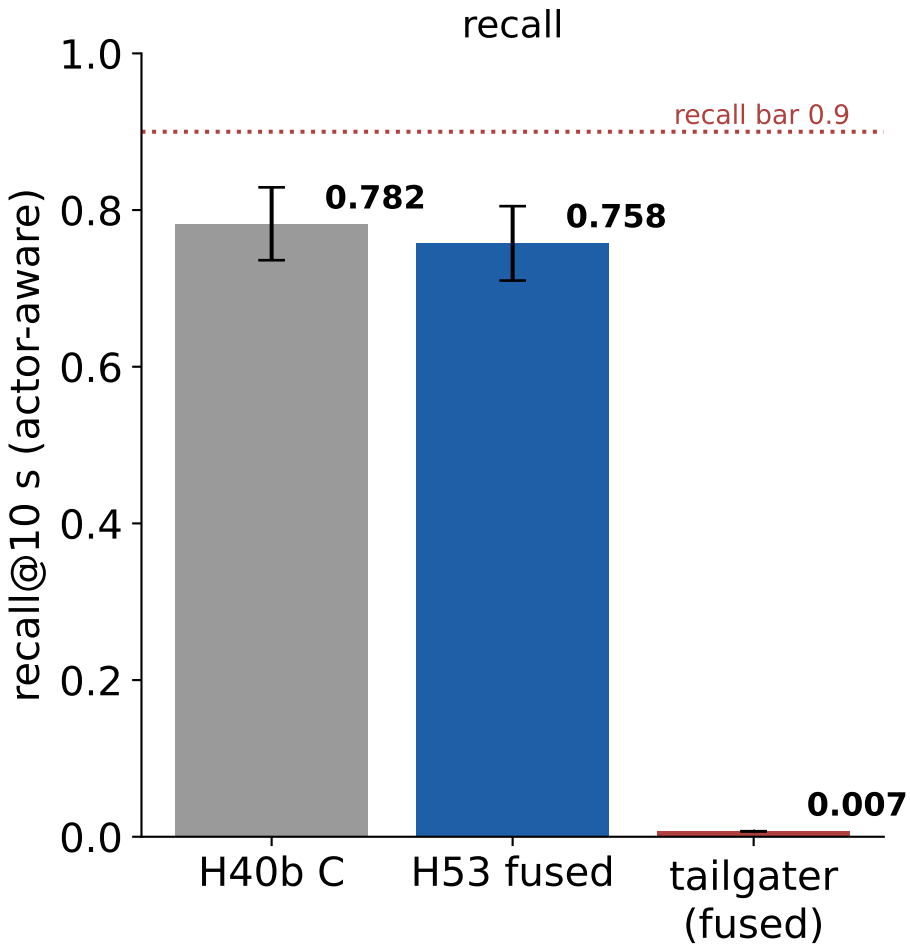
8 · Drift monitor: textbook tests fire all the time



H34 (9 cams × 12 injected shifts, 1-s windows): PSI and KS fire on 100 % of stationary windows = 1,440 FA/24 cam-h; calibrated z detects 6/12 at 9.2 FA/24 cam-h.
Embedding arm: 0 null fires but 0/12 detected. Verdict INCONCLUSIVE under the pre-registered rule.
1,440 assumes one evaluation per minute (1,440 min/day); at the 1-s window rate it would be 60× higher.

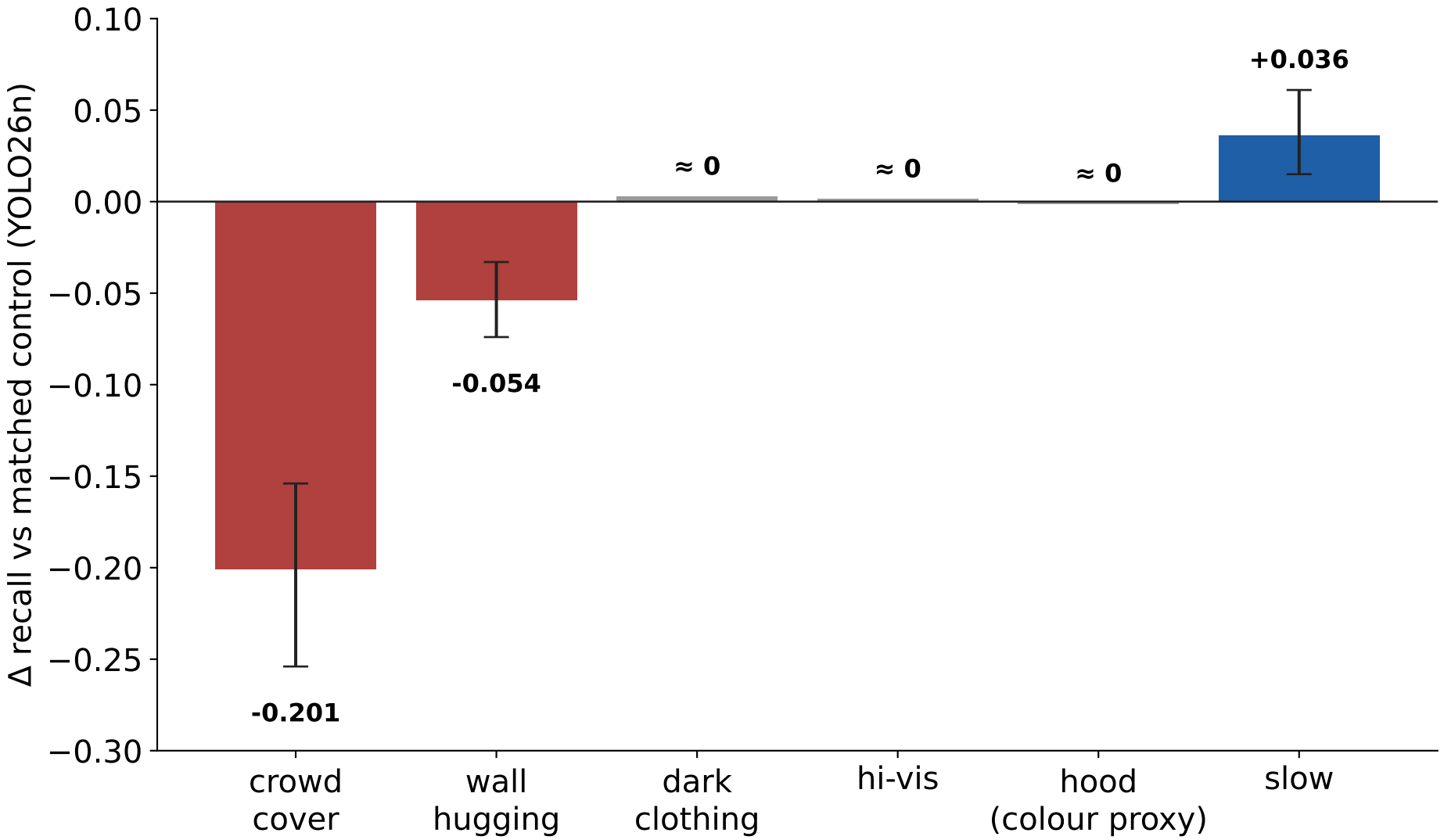
9 · Line-crossing: rule → twins → fusion





H40 (old unfenced worlds, assumed tracker noise): naive 70.96 → gated 4.20 FA/cam-h; recall C 0.100.
H40b (fenced worlds, 3,002 clean cam-h): arm C non-twin FA 0 (rule of three ≤ 0.001), but authorised twins (patrol + maintenance) 0.488 FA/cam-h; recall@10 0.782.
H53 late fusion (gate log + BLE tag + PIDS): twins → 0.0047 FA/cam-h (99 % removed), recall 0.782 → 0.758. ALL sensors simulated, parameters assumed, matched on TRUE positions.
A tailgater defeats the gate log: tailgate recall 0.007 combined.

10 · Evasion: crowd cover hurts, clothing does not (H47)



H47: YOLO26n zero-shot recall vs matched control (0.948), 2 cams × 2–3 seeds, 30 s clips; clip-cluster 95 % CIs (AUDIT_H47): crowd_cover −0.201 [−0.254, −0.154], wall_hugging −0.054 [−0.074, −0.033] (43 % fewer ≥ 20 px frames), slow +0.036 [+0.015, +0.061]; dark / hi-vis / hood (colour proxy) |Δ| ≤ 0.003. Not met (1/6). Only 4 clips per technique, so min clip-level p = 1/16.
UNTESTED (no CC0 asset): umbrella, carried object, crouch/crawl, face-away.